RAFAEL ROHR

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EDUCATION

Duke University **Aug 2024 – May 2028** | Durham, NC (USA)
BS: Electrical & Computer Engineering; Math | **Minor:** Finance | **Relevant Coursework:** Probability, Financial Markets, Digital Systems, Signals and Systems, Mathematical Finance (**G**rad), Adv. Computer Architecture I (**G**)
GPA: 3.90 | Dean's List with Distinction (Spring 2025), Dean's List (Fall 2025)

Honors and Awards: Biogenesis International Biomedicine Olympiad: Gold (2022, Molecular and Cellular Biology category); Gold (2022, Anatomy and Physiology category); Bronze (2022, Pharmacology and Toxicology category), Brazilian Chemistry Olympiad (**OBQ**): Bronze (2023); Bronze (2022), Brazilian Physics Olympiad (**OBF**): Silver (2023), Brazilian National Sciences Olympiad (**ONC**): Gold (2023); Gold (2022), Brazilian Astronomy and Astronautics Olympiad (**OBA**): Gold (2022), Brazilian Robotics Olympiad (**OBR**): Silver (2022)

SKILLS & INTERESTS

Skills: Java, TypeScript, JavaScript, SQL, C, Python, Verilog, HTML, CSS, Onshape, Linux, API Development, Assembly, Git, Vivado, FPGA, React
Technologies: AWS (S3, Glue, SageMaker), GTKWave, Postman, Docker Compose, Logisim, Arduino, Jira, Figma, Salesforce CRM, Tessent Shell, Datadog, Gem
Languages: English (Fluent), Portuguese (Fluent), Spanish (C1), French (B1)
Interests: Soccer (Chelsea, Madrid, and Brazil), Chess, Philosophy, Poetry Writing, Poker & Strategy Games, Weightlifting, College Basketball

INTERNSHIPS

Itaú Unibanco (Private Bank) | *AI and Software Engineering Intern* | Read More **May 2026 – Aug 2026** | Sao Paulo, SP (Brazil)
- Built an AI investment intelligence platform that detected portfolio anomalies and risk events, and generated executive reports for proactive portfolio monitoring
- Architected end-to-end AWS S3, Glue, SageMaker, and LangGraph pipeline to automate analysis and reports, scaling an analytical workflow into a production process
- Created GPIS' centralized product platform integrating Microsoft Stack and Amazon QuickSight, creating a single source of truth, improving visibility and execution
- Developed an AI workflow to flag discretionary portfolio management high-propensity clients, easing bankers' operations, boosting data-driven client engagement

StoneCo | *Software Engineer and Product Manager Intern* | Read More **May 2025 – Aug 2025** | Sao Paulo, SP (Brazil)
- Acted as PM and full-stack SWE in Brazil's largest fintech, building and deploying a feature that saved chat attendants time in each call by automating their workflow
- Saved 49 seconds per session (savings of ~R$1M (US$200K) annually) by implementing AI-driven prompt engineering with secure handling of sensitive user data
- Partnered with product, design, AI, and R&D teams to integrate Salesforce CRM, AWS S3, and Twilio to deliver a secure, scalable, and compliant fintech solution
- Created a fine-tuned ChatGPT model, conducted internal research to achieve accurate humanistic output, and used the custom model in a TypeScript API endpoint

CAMPUS EXPERIENCE

Duke Department of Computer Science | *Head Teaching Assistant for Data Structures and Algorithms* **Aug 2025 – Present** | Durham, NC (USA)
- Lead weekly class discussions for ~30 students, fostering critical thinking, reinforcing concepts in data structures, algorithms, recursion, and asymptotic analysis
- Hold office hours to individually support assignments, teaching debugging techniques, and guiding students to develop strong problem-solving and coding practices
- Assist with grading by providing thorough feedback on content, covering abstract data types, trees, graphs, and introductory AI/ML algorithms implemented in Java

Duke Department of Electrical & Computer Engineering | *Teaching Assistant for Fundamentals of ECE* **Aug 2026 – Present** | Durham, NC (USA)
- Hold office hours supporting ~10 students weekly with circuit analysis, analog electronics, frequency response filtering, and digital logic concepts
- Grade weekly assignments and provide detailed technical feedback, reinforcing rigorous problem-solving and clear engineering communication

Duke Annual Fund (Phonathon) | *Alumni Relations and Outreach Representative* **Jan 2026 – Jun 2026** | Durham, NC (USA)
- Secured ~$2,500 in contributions across 15+ fundraising sessions by building relationships with the Duke community through persuasive, high-impact communication
- Managed sensitive donor information, including payment and credit card data, with strict adherence to confidentiality, accuracy, and university data security protocols
- Developed stakeholder management, sales, and data-handling skills, engaging diverse alumni profiles in a fast-paced, metrics-driven outreach environment

PROJECTS & RESEARCH

Duke & Siemens EDA | *Design for Testability (DFT) researcher for pipelined RISC-V processor* | Read More **Jan 2026 – May 2026** | Durham, NC (USA)
- Collaborated with Siemens EDA engineers to implement DFT methods using Tessent Shell on a CPU, developing scan insertion, ATPG, and fault analysis scripts
- Achieved 95% stuck-at and 90.5% transition fault coverage, gaining expertise in digital design verification, semiconductor testing, and hardware reliability engineering

Spell-it-Out | *FPGA Engineer* | Read More **Apr 2026 – May 2026** | Durham, NC (USA)
- Built a whiteboard-writing robotic system integrating a self-designed Verilog 5-stage pipelined MIPS CPU, FPGA-controlled motors, servos, and PS2 keyboard input
- Developed assembly and memory-mapped FPGA interfaces for real-time text-to-motion execution, synchronized 3-axis control, and automated character formatting
- Verified and debugged RTL using Verilog testbenches, waveform analysis, and a digital oscilloscope, achieving reliable autonomous writing of all 26 letters

Christensen Family Center for Innovation | *Researcher* | Read More **Aug 2025 – May 2026** | Durham, NC (USA)
- Developed a full-stack platform using React, TypeScript, and Python to organize technical literature and accelerate evaluation of emerging technologies
- Engineered backend and AI API workflows to analyze uploaded documents, synthesize research insights, to generate context-specific commercialization opportunities
- Researched AI-driven methods to identify alternative use cases for emerging engineering technologies, emphasizing innovation and commercialization potential
- Conducted market analysis and customer discovery to assess use-case feasibility, transforming qualitative feedback into product features and engineering requirements

LEADERSHIP & EXTRACURRICULAR

Brazilian Student Association (BRASA) | *President* **Sep 2024 – Present** | Durham, NC (USA)
- Plan and execute biweekly events, promoting a tight-knit Brazilian community and spreading Brazilian culture to foreign students, engaging 100+ students
- Secured over $20k in funding, led a 25-member team, and mentored high school students on Duke applications, corrected 20+ essays, and hosted application panels
- Developed professional partnerships with companies such as EMS, Itaú, and StoneCo, and brought Alumni from Google, Microsoft, and Bain to info sessions
- Grew membership 23%, reached 45K monthly Instagram accesses, and earned BRASA's Most Opportunity-Generating Chapter award